Inteligență artificială centrată pe om pentru gestionarea energiei
Intelligence artificielle centrée sur l'humain pour la gestion de l'énergie
Human centered AI for energy management

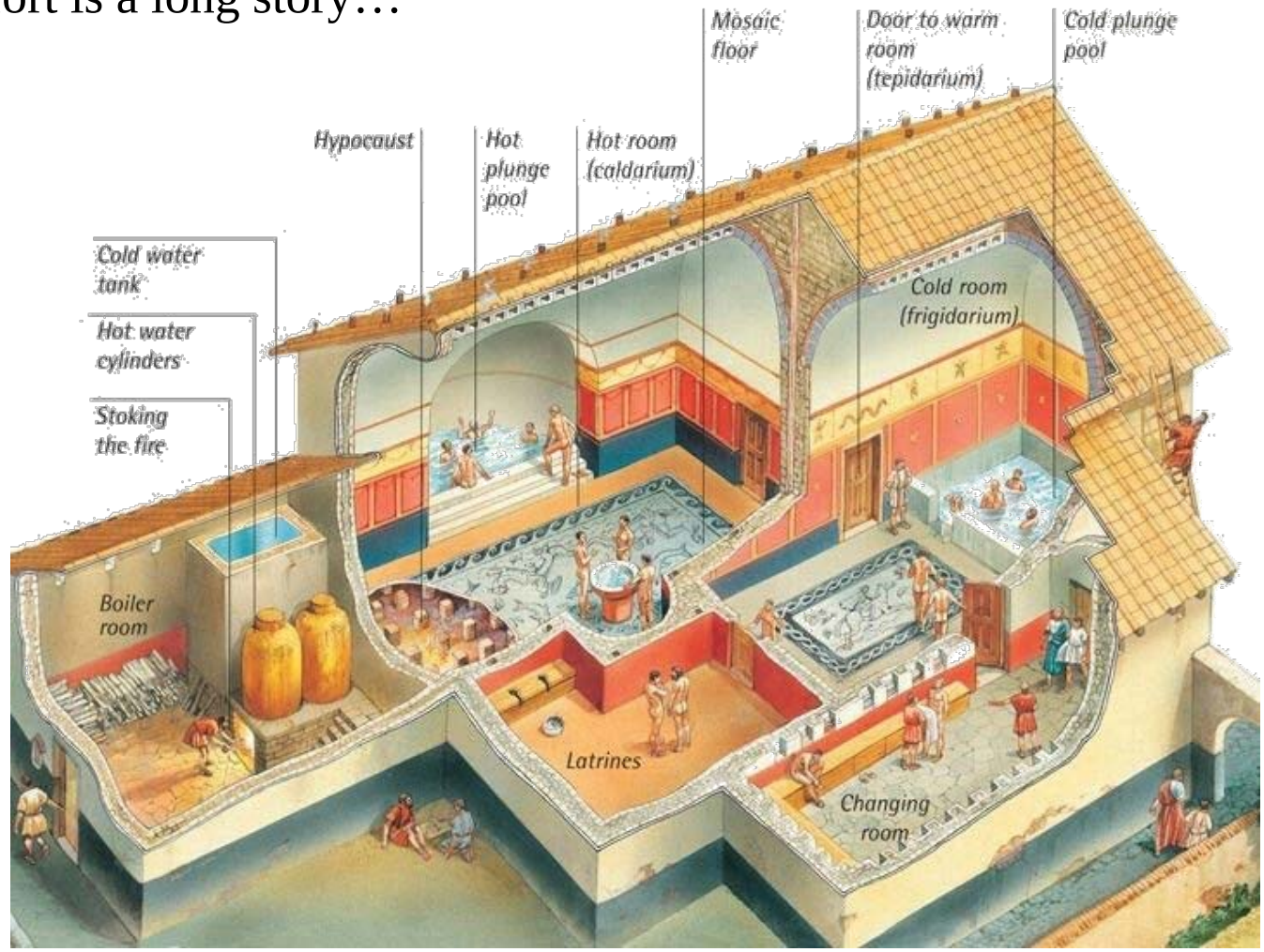
Chapter 2 – OCCUPANTS' COMFORT AND HEALTH

- Thermal and visual comfort: temperature, humidity, ventilation, lighting.
- Thermal comfort indices: PMV (Predicted Mean Vote) and PPD (Predicted Percentage Dissatisfied); calculation and interpretation according to ISO 7730.
- Indoor air quality: concentration of CO_2, volatile organic compounds (VOC) and suspended particles as indicators of occupant health and productivity.
- Reference standards and regulations: EN 15251, ASHRAE 55.



[Image generated with Nano Banana]

Human comfort is a long story…



[https://ro.pinterest.com/pin/502784745892885384/]

2.1. Thermal and visual comfort

2.1.1 General considerations on atmospheric (moist) air

Atmospheric air = a mixture of dry air and water vapor = moist air
The **dry air** is treated as if it was a pure and stable component. The approximate composition of dry air by volume percent (component volume / dry air volume) is the following:

Nitrogen	78.08
Oxygen	20.95
Argon	0.93
Carbon dioxide	0.03
Other gases such as neon, sulfur dioxide, etc.	0.01

The amount of **water vapor** present in moist air at a temperature range of -17.8°C to 37.8°C varies from 0.05 to 3% by mass (mass of water vapor / mass of moist air).
Water vapor is lighter than air => a cloud in the sky floats in the air.

In air-conditioning applications, the water vapors in the air (moist air) are only in a gaseous phase (superheated or at saturation).

Psychrometrics is the study of the thermodynamic properties of moist air.

The overall mixture and each mixture component behave as **ideal gases** => **very simple laws for modelling.**

Equation of State of an Ideal Gas

$$pV = mRT$$

p = absolute pressure of atmospheric air [Pa]
V = moist air volume [m^3]
m= mass of moist air [kg]
R = 287 J/kgK air specific constant
T = absolute temperature of moist air [K]

Other forms:

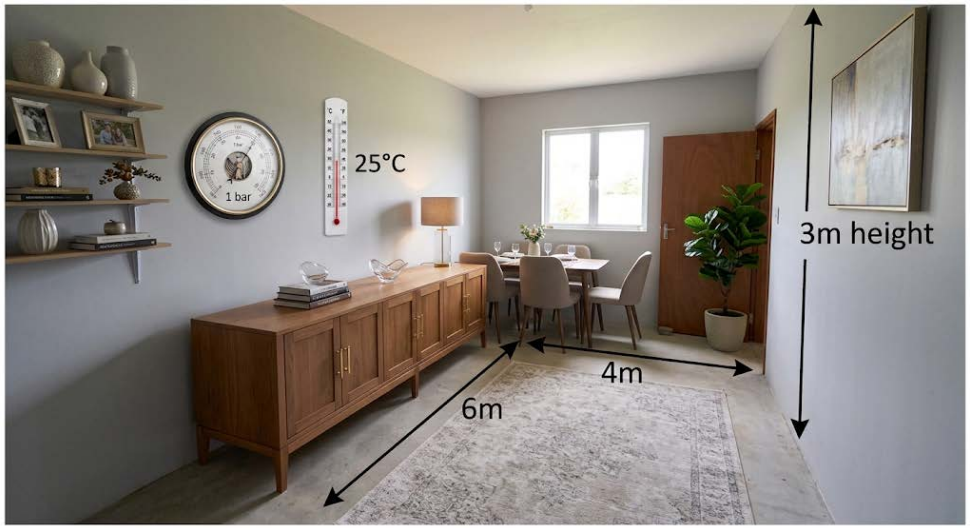
$$pv = RT \;\; ; \; v = V/m$$

$$p = \rho RT \;\; ; \; \rho = m/V$$

ρ = air density [kg/m^3]



LECTURE CHALLENGE – mass and density of air in the room?



1 bar = 10^5 Pa; 25°C = (25+273.15) K

$$m = \frac{pV}{RT} =$$

$$\rho = \frac{p}{RT} = \frac{m}{V} =$$

Answers:

$$m = \frac{pV}{RT} = \frac{10^5 * (6 * 4 * 3)}{287 * (25 + 273.15)} = 84.14\ kg$$

$$\rho = \frac{p}{RT} = \frac{m}{V} = \frac{84.14}{72} = 1.16\ \frac{kg}{m^3}$$

Properties of Moist Air

Although air contaminants may seriously affect the health of occupants of the air conditioned space, they have little effect on the thermodynamic properties of moist air since their mass concentration is low. For simplicity, moist air is always considered as a **binary mixture of dry air and water vapor** during the analysis and calculation of its properties [1].

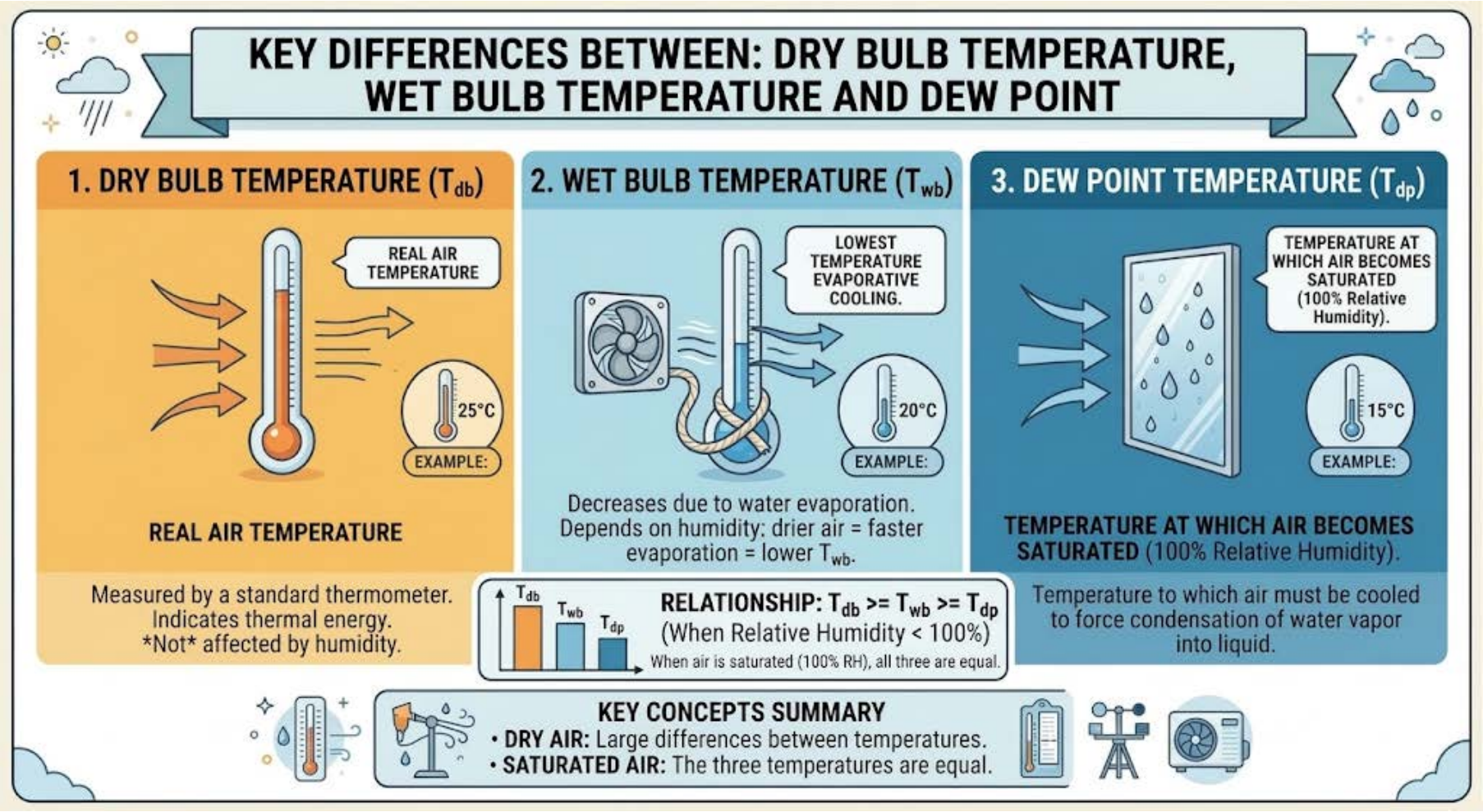
- **Moist air pressure** $p = p_a + p_v$ [Dalton's law]

  p_a, p_v partial pressure of dry air and water vapor, respectively

- **Moist air mass** $m = m_a + m_v$ [principle of mass conservation]

  m_a, m_v mass of dry air and water vapor, respectively

➢ Dry bulb temperature, wet bulb temperature and dew point temperature



[Image generated with Nano Banana]

- **Humidity Ratio (or moisture content)** $x = \frac{m_v}{m_a} \left[\frac{kg\ water\ vapor}{kg\ dry\ air}\right]$

  Average annual value for Romania: *x*=10 g/kg.

  Other useful derived equation: $x = \frac{m_v}{m_a} = 0.622 \frac{p_v}{p - p_v}$

- **Relative Humidity** $\varphi = \frac{p_v}{p_{sat}(T)}$

  $p_{sat}(T)$ = saturation pressure of water at moist air temperature *T*

  This is a relative measure that shows how close, or far, is a state of the moist air from that of saturation at a specific temperature *T*.

- **Enthalpy of moist air** $h = m_a h_a + m_v h_v = m_a(h_a + x h_v) \approx t + x(2500 + 1.84t) \left[\frac{kJ}{kg}\right]$, t[°C]



LECTURE CHALLENGE – compare a summer day to a winter day in Romania

Summer day in Romania	**Winter day in Romania**
p = 1 bar	p = 1 bar
t = 30°C	t = 0°C
φ = 60%	φ = 85%
$p_{sat}(t)=$	$p_{sat}(t)=$
$p_v = \varphi * p_{sat}(t)=$	$p_v =$
$x = 0.622 \frac{p_v}{p-p_v} =$	$x =$
$t_{dp} = t_{sat}\,(p_v)$	$t_{dp} =$
$h= t + x(2500 + 1.84t) =$	$h=$

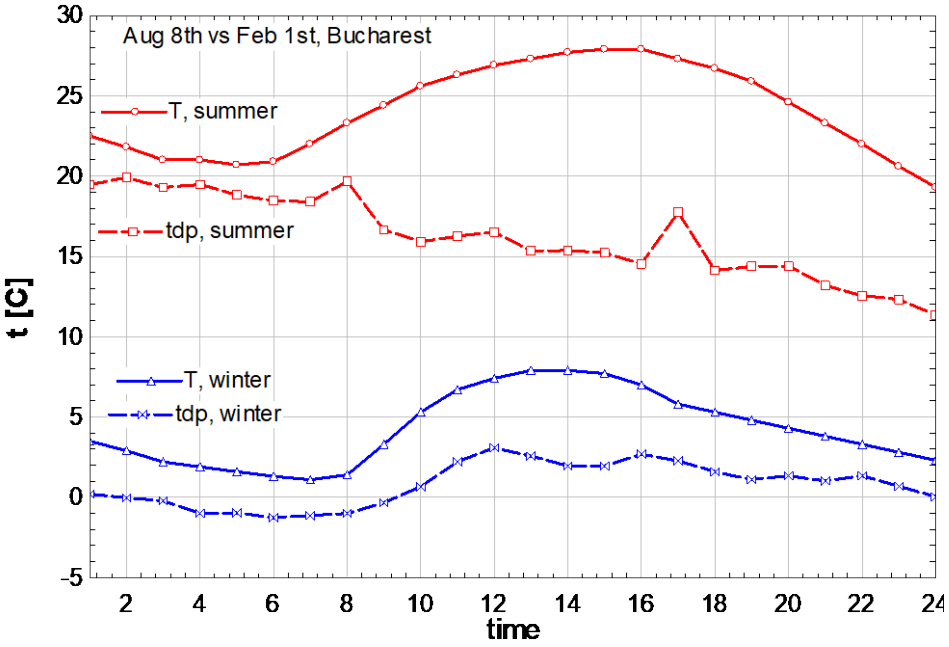
Known data for water at saturation:

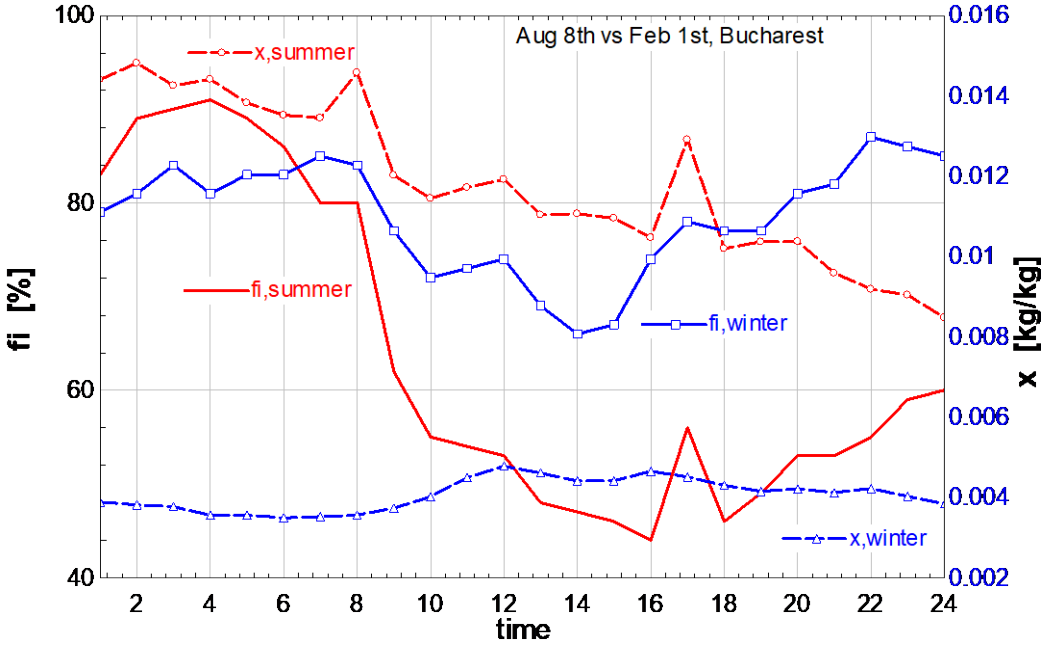
t [°C]	**-6**	**-4**	**-2**	**0**	**2**	**4**	**6**	**8**	**10**	**12**	**14**	**16**
p_{sat} **[Pa]**	390.9	454.9	527.9	611.2	706	813.5	935.3	1073	1228	1403	1599	1819
t [°C]	**18**	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**	**38**	**40**
p_{sat} **[Pa]**	2065	2339	2645	2986	3364	3783	4247	4760	5325	5948	6633	7385

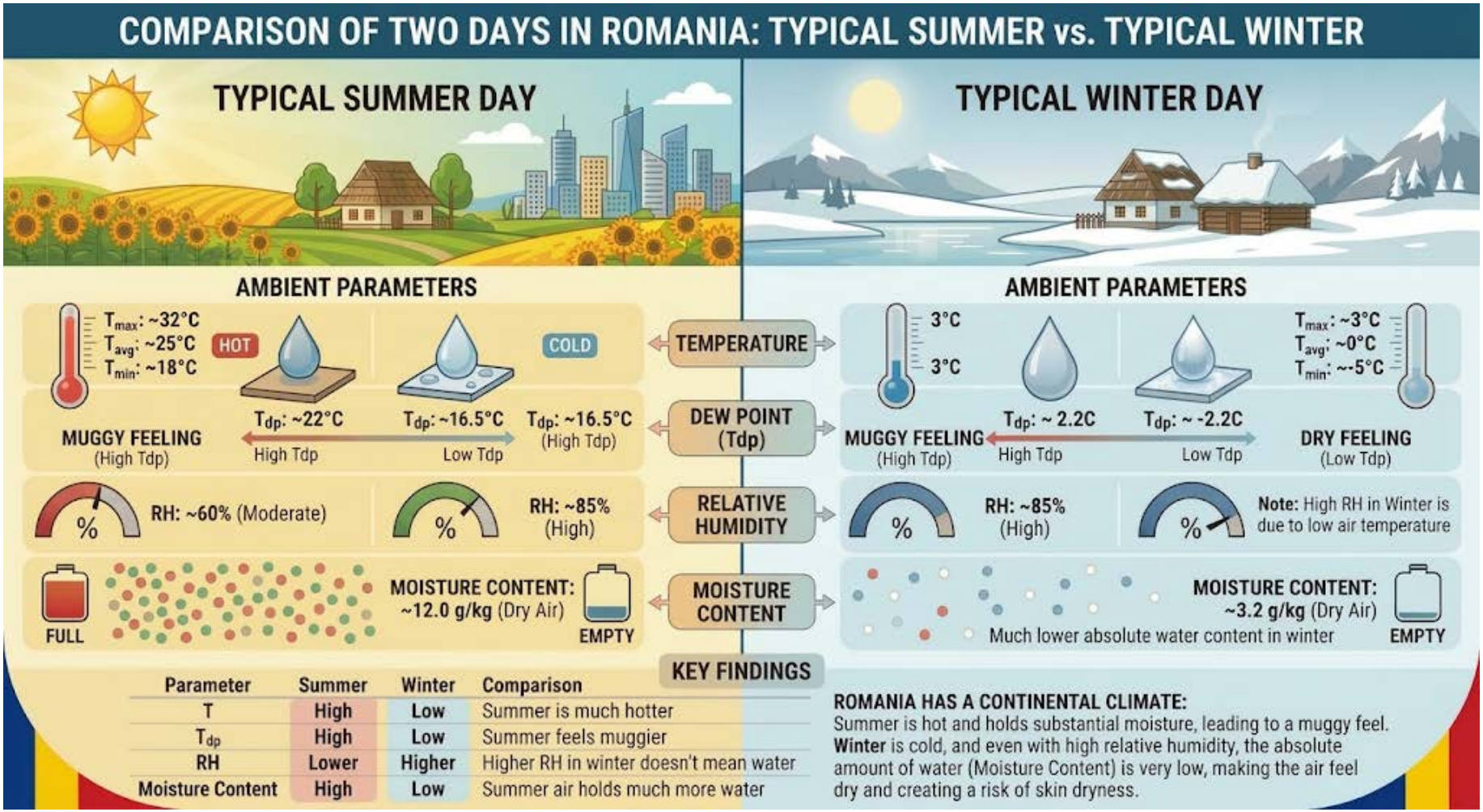
Individual work to build together an entire day simulation!

Bucharest – Meteornorm data, August 8th versus February 1st

Summer day			Winter day		
Time, UTC	**t [°C]**	**φ [%]**	**Time**	**t [°C]**	**φ [%]**
1	22.5	83	1	3.5	79
2	21.8	89	2	2.9	81
3	21	90	3	2.2	84
4	21	91	4	1.9	81
5	20.7	89	5	1.6	83
6	20.9	86	6	1.3	83
7	22	80	7	1.1	85
8	23.3	80	8	1.4	84
9	24.4	62	9	3.3	77
10	25.6	55	10	5.3	72
11	26.3	54	11	6.7	73
12	26.9	53	12	7.4	74
13	27.3	48	13	7.9	69
14	27.7	47	14	7.9	66
15	27.9	46	15	7.7	67
16	27.9	44	16	7	74
17	27.3	56	17	5.8	78
18	26.7	46	18	5.3	77
19	25.9	49	19	4.8	77
20	24.6	53	20	4.3	81
21	23.3	53	21	3.8	82
22	22	55	22	3.3	87
23	20.6	59	23	2.8	86
24	19.3	60	24	2.3	85


Aug 8th vs Feb 1st, Bucharest
T, summer
tdp, summer
T, winter
tdp, winter
t [C]
time
30
25
20
15
10
5
0
-5
2
4
6
8
10
12
14
16
18
20
22
24


Aug 8th vs Feb 1st, Bucharest
x,summer
fi,summer
fi,winter
x,winter
fi [%]
x [kg/kg]
time
100
80
60
40
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
2
4
6
8
10
12
14
16
18
20
22
24



[Image generated with Nano Banana]


GREAT TEAM!
WE LEARN EVERY DAY!
WOW, LOOK WHAT YOU'VE LEARNED!
WELL DONE!
STORIES
STORIES
EXPERIMENTS
SPACE

2.1.2 Thermal comfort

While psychrometrics strictly defines the physical properties of the air — such as dry-bulb temperature, moisture content and relative humidity — **human thermal comfort is fundamentally a dynamic heat transfer balance between the human body and its immediate environment**.
According to established stanrdards and regulations, this delicate balance is **governed not only by the four environmental factors (air temperature, mean radiant temperature, air velocity, and humidity)** but also by two highly variable personal factors: **metabolic rate and clothing insulation**.

In the context of human-centered AI and smart energy management, we must shift our focus from maintaining static psychrometric setpoints to optimizing dynamic comfort indices, such as the Predicted Mean Vote (PMV) and Predicted Percentage Dissatisfied (PPD). By integrating these comprehensive metrics into dynamic thermal simulations, **AI algorithms can continuously adapt the indoor environment to satisfy the complex physiological needs of the occupants while minimizing the building's overall energy consumption**.

Indoor air temperature and humidity should follow the ASHRAE comfort recommendations - ASHRAE 55 (Thermal Environmental Conditions for Human Occupancy).

According to the ASHRAE 55 standard, achieving thermal comfort requires **maintaining indoor conditions within a specific psychrometric zone, accounting for seasonal variations in clothing insulation and metabolic rates**.

- **Operative Temperature:** The recommended operative temperature typically ranges from **20°C to 24°C** during the winter (heating season, assuming standard winter clothing of ~1.0 clo) and from **23°C to 26°C** during the summer (cooling season, assuming lighter clothing of ~0.5 clo).

- **Humidity Control:** ASHRAE 55 imposes a strict upper limit defined by a maximum humidity ratio of **0.012 kg of water vapor per kg of dry air** (which approximately corresponds to **60% relative humidity** at standard indoor temperatures). While the standard does not specify a strict lower limit for thermal comfort alone, maintaining indoor relative humidity between **30% and 60%** is widely accepted as optimal to prevent respiratory discomfort and microbial growth.

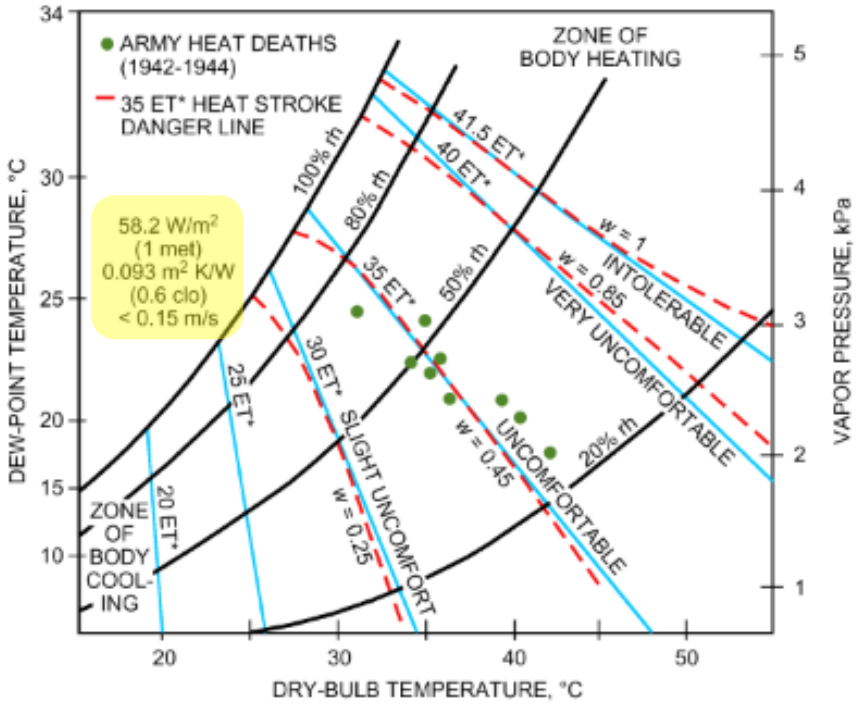


Fig. 2 Isotherms for Comfort, Discomfort, Physiological Strain, Effective Temperature (ET*), and Heat Stroke Danger Threshold [2]

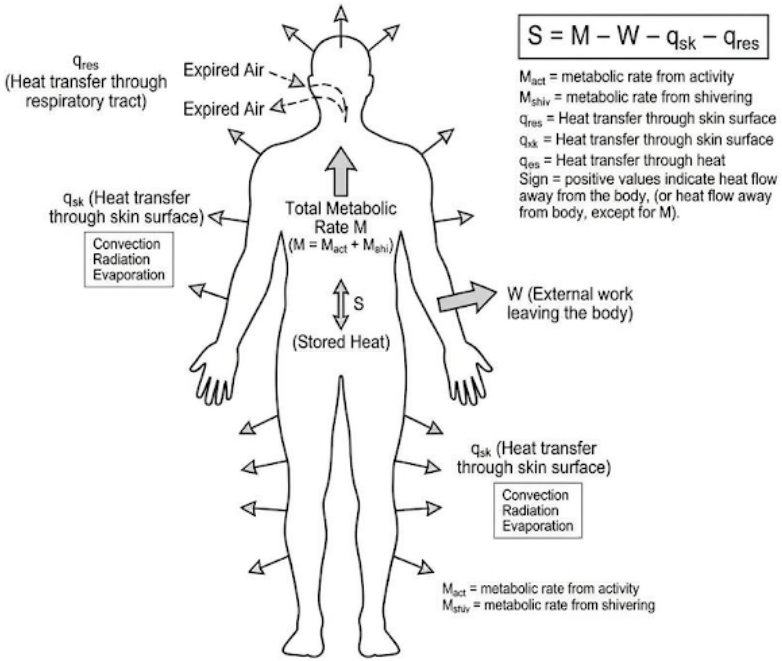
Metabolic rate

A resting adult produces about **100 W** of heat. It is more convenient to express the heat released by skin per m^2 of skin.

An **average European male** has a skin surface area of about **1.8 m^2**.

By comparison, a female European has an average surface area of 1.6 m^2.

100 W / 1.7 m^2=> **58 W/m^2 , called 1 met.**



Body thermal balance

⇨ body temperature increases or decreases

⇨ need for adjusting operative temperature

$$t_{sk} = 35.7 - 0.0275(M - W)$$

M = metabolic rate

W = mechanical work performed

At higher activity levels, sweat loss increases and mean skin temperature decreases.

Body Surface Area

The terms in Equation (1) have units of power per unit area and refer to the surface area of the nude body. The most useful measure of nude body surface area, originally proposed by DuBois and DuBois (1916), is described by

$$A_D = 0.202 m^{0.425} l^{0.725} \quad (4)$$

where

A_D = DuBois surface area, m^2
m = mass, kg
l = height, m

A correction factor $f_{cl} = A_{cl}/A_D$ must be applied to the heat transfer terms from the skin (C, R, and E_{sk}) to account for the actual surface area A_{cl} of the clothed body. Table 7 presents f_{cl} values for various clothing ensembles. For a 1.73 m tall, 70 kg man, $A_D = 1.8$ m^2. All terms in the basic heat balance equations are expressed per unit DuBois surface area.

[2]

Table 7 Typical Insulation and Permeability Values for Western Clothing Ensembles

Ensemble Description[a]	I_{cl}, clo	I_t,[b] clo	f_{cl}	i_{cl}	i_m[b]
Walking shorts, short-sleeved shirt	0.36	1.02	1.10	0.34	0.42
Trousers, short-sleeved shirt	0.57	1.20	1.15	0.36	0.43
Trousers, long-sleeved shirt	0.61	1.21	1.20	0.41	0.45
Same as above, plus suit jacket	0.96	1.54	1.23		
Same as above, plus vest and T-shirt	1.14	1.69	1.32	0.32	0.37
Trousers, long-sleeved shirt, long-sleeved sweater, T-shirt	1.01	1.56	1.28		
Same as above, plus suit jacket and long underwear bottoms	1.30	1.83	1.33		
Sweat pants, sweat shirt	0.74	1.35	1.19	0.41	0.45
Long-sleeved pajama top, long pajama trousers, short 3/4 sleeved robe, slippers (no socks)	0.96	1.50	1.32	0.37	0.41
Knee-length skirt, short-sleeved shirt, panty hose, sandals	0.54	1.10	1.26		
Knee-length skirt, long-sleeved shirt, full slip, panty hose	0.67	1.22	1.29		
Knee-length skirt, long-sleeved shirt, half slip, panty hose, long-sleeved sweater	1.10	1.59	1.46		
Same as above, replace sweater with suit jacket	1.04	1.60	1.30	0.35	0.40
Ankle-length skirt, long-sleeved shirt, suit jacket, panty hose	1.10	1.59	1.46		
Long-sleeved coveralls, T-shirt	0.72	1.30	1.23		
Overalls, long-sleeved shirt, T-shirt	0.89	1.46	1.27	0.35	0.40
Insulated coveralls, long-sleeved thermal underwear, long underwear bottoms	1.37	1.94	1.26	0.35	0.39

Source: From McCullough and Jones (1984) and McCullough et al. (1989).
[a]All ensembles include shoes and briefs or panties. All ensembles except those with panty hose include socks unless otherwise noted.
[b]For $t_r = t_a$ and air velocity less than 0.2 m/s ($I_a = 0.72$ clo and $i_m = 0.48$ when nude).
1 clo = 0.155 ($m^2 \cdot K/W$).

Table 4 Typical Metabolic Heat Generation for Various Activities

	W/m²	met*
Resting		
Sleeping	40	0.7
Reclining	45	0.8
Seated, quiet	60	1.0
Standing, relaxed	70	1.2
Walking (on level surface)		
3.2 km/h (0.9 m/s)	115	2.0
4.3 km/h (1.2 m/s)	150	2.6
6.4 km/h (1.8 m/s)	220	3.8
Office Activities		
Reading, seated	55	1.0
Writing	60	1.0
Typing	65	1.1
Filing, seated	70	1.2
Filing, standing	80	1.4
Walking about	100	1.7
Lifting/packing	120	2.1
Driving/Flying		
Car	60 to 115	1.0 to 2.0
Aircraft, routine	70	1.2
Aircraft, instrument landing	105	1.8
Aircraft, combat	140	2.4
Heavy vehicle	185	3.2
Miscellaneous Occupational Activities		
Cooking	95 to 115	1.6 to 2.0
Housecleaning	115 to 200	2.0 to 3.4
Seated, heavy limb movement	130	2.2
Machine work		
sawing (table saw)	105	1.8
light (electrical industry)	115 to 140	2.0 to 2.4
heavy	235	4.0
Handling 50 kg bags	235	4.0
Pick and shovel work	235 to 280	4.0 to 4.8
Miscellaneous Leisure Activities		
Dancing, social	140 to 255	2.4 to 4.4
Calisthenics/exercise	175 to 235	3.0 to 4.0
Tennis, singles	210 to 270	3.6 to 4.0
Basketball	290 to 440	5.0 to 7.6
Wrestling, competitive	410 to 505	7.0 to 8.7

Sources: Compiled from various sources. For additional information, see Buskirk (1960), Passmore and Durnin (1967), and Webb (1964).

*1 met = 58.1 W/m²

[2]

Going deeper inside metabolic rate (for the curious minds)

Estimating metabolic rates is difficult as it depends on how well the activity is defined.

Usually, for engineering applications, for M<1.5 met the accuracy is very good. For M>3met, a task is poorly defined and the values may be in error by as much as ±50% for a given application.

The rate of metabolic heat produced by the body is most accurately measured by the rate of respiratory oxygen consumption and carbon dioxide production. An empirical equation for metabolic rate is given by **Nishi** (1981):

$$M = \frac{21(0.23\,\mathrm{RQ} + 0.77)V_{O_2}}{A_D}\left[\frac{W}{m^2}\right]$$

RQ [-] = respiratory quotient; $RQ = \frac{n_{M,CO_2}}{n_{M,O_2}} = \frac{V_{CO_2}}{V_{O_2}}$ ratio of exhaled CO_2 to inhaled O_2.

The RQ value indicates exactly what type of "fuel" the human body is burning at a given time to generate energy (and consequently, heat):

- **RQ = 1.0:** The body is metabolizing exclusively **carbohydrates** (for every oxygen molecule consumed, exactly one CO_2 molecule is produced).
- **RQ ≈ 0.7:** The body is predominantly metabolizing **fats** (lipids), a process that requires more oxygen.
- **RQ ≈ 0.8 - 0.85:** This is the typical average value for a person on a mixed diet or when metabolizing proteins.

A good estimate for the average adult is RQ = 0.83 for light or sedentary activities (M < 1.5 met), increasing proportionately to RQ = 1.0 for extremely heavy exertion (M = 5 met).

V_{O_2} [mL/s]:

Table 5 Heart Rate and Oxygen Consumption at Different Activity Levels

Level of Exertion	Heart Rate, bpm	Oxygen Consumed, mL/s
Light work	<90	<8
Moderate work	90 to 110	8 to 16
Heavy work	110 to 130	16 to 24
Very heavy work	130 to 150	24 to 32
Extremely heavy work	150 to 170	>32

Source: Astrand and Rodahl (1977).

[2]

M varies not only based on physical exertion (sitting vs. walking) but also on internal chemical processes. By knowing exactly how much energy the body produces (through RQ and oxygen consumption), one can feed highly precise data into the artificial intelligence models that control HVAC systems to optimize and maintain thermal comfort.

Example:

	Light work	Heavy work
RQ	0.83	0.9
V_{O_2} [mL/s]	7	20
A_D $[m^2]$	1.8	1.8
$M = \frac{21(0.23\ \mathrm{RQ} + 0.77)V_{O_2}}{A_D}\ \left[\frac{W}{m^2}\right]$	78.47	227.96
met = M[W/m^2] / 58	1.35	3.93

Operative Temperature = weighted average between the indoor air temperature and the mean radiant temperature of the room's surfaces; it is **a much more accurate indicator of human thermal comfort perception than simple air temperature**:

$$t_o = \frac{h_c t_a + h_r t_{mr}}{h_c + h_r}$$

where:

- t_a = indoor air temperature (°C)
- t_{mr} = mean radiant temperature of the walls and surrounding surfaces (°C)
- h_c = convective heat transfer coefficient at the surface of the human body (W/m^2K)
- h_r = linearized radiative heat transfer coefficient (W/m^2K)

Table 6 Equations for Convection Heat Transfer Coefficients

Equation	Limits	Condition	Remarks/Sources
$h_c = 8.3V^{0.6}$ $h_c = 3.1$	$0.2 < V < 4.0$ $0 < V < 0.2$	Seated with moving air	Mitchell (1974)
$h_c = 2.7 + 8.7V^{0.67}$ $h_c = 5.1$	$0.15 < V < 1.5$ $0 < V < 0.15$	Reclining with moving air	Colin and Houdas (1967)
$h_c = 8.6V^{0.53}$	$0.5 < V < 2.0$	Walking in still air	V is walking speed (Nishi and Gagge 1970)
$h_c = 5.7(M - 0.8)^{0.39}$	$1.1 < M < 3.0$	Active in still air	Gagge et al. (1976)
$h_c = 6.5V^{0.39}$	$0.5 < V < 2.0$	Walking on treadmill in still air	V is treadmill speed (Nishi and Gagge 1970)
$h_c = 14.8V^{0.69}$ $h_c = 4.0$	$0.15 < V < 1.5$ $0 < V < 0.15$	Standing person in moving air	Developed from data presented by Seppänen et al. (1972)

Note: h_c in W/(m² · K), V in m/s, and M in met, where 1 met = 58.1 W/m².

$$h_c = \begin{cases} 2.38(t_{cl} - t_a)^{0.25} & 2.38(t_{cl} - t_a)^{0.25} > 12.1\sqrt{V} \\ 12.1\sqrt{V} & 2.38(t_{cl} - t_a)^{0.25} < 12.1\sqrt{V} \end{cases}$$

$$f_{cl} = \begin{cases} 1.0 + 0.2I_{cl} & I_{cl} < 0.5 \text{ clo} \\ 1.05 + 0.1I_{cl} & I_{cl} > 0.5 \text{ clo} \end{cases}$$

[2]

$$h_r = 4\varepsilon\sigma \frac{A_r}{A_D}\left(273.2 + \frac{t_{cl} + t_{mr}}{2}\right)^3$$

ε = average emissivity of clothing or body surface

σ = Stefan-Boltzmann constant, 5.67×10^{-8} W/(m²·K⁴)

A_r = effective radiation area of body, m²

$\frac{A_r}{A_D} = 0.70$ for a sitting person and 0.73 for a standing person (Fanger 1967)

Mean radiant temperature:

- for a standing person

$$t_{mr} = \frac{0.08\left[t_{pr}(\text{up}) + t_{pr}(\text{down})\right] + 0.23[t_{pr}(\text{right}) + t_{pr}(\text{left})] + 0.35\left[t_{pr}(\text{front}) + t_{pr}(\text{back})\right]}{[2(0.08 + 0.23 + 0.35)]}$$

- for a seated person

$$t_{mr} = \frac{0.18\left[t_{pr}(\text{up}) + t_{pr}(\text{down})\right] + 0.22[t_{pr}(\text{right}) + t_{pr}(\text{left})] + 0.3\left[t_{pr}(\text{front}) + t_{pr}(\text{back})\right]}{[2(0.18 + 0.22 + 0.3)]}$$

t_{pr} = plane radiant temperature

CONDITIONS FOR THERMAL COMFORT

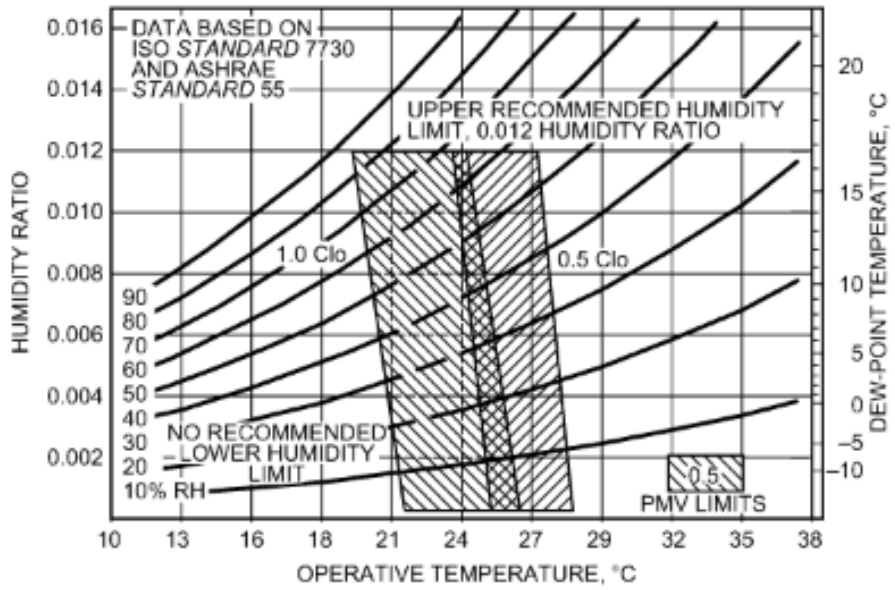
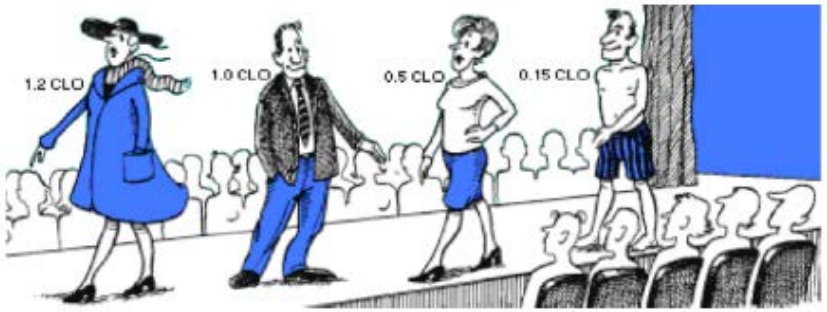


Fig. 5 ASHRAE Summer and Winter Comfort Zones
[Acceptable ranges of operative temperature and humidity with air speed ≤ 0.2 m/s for people wearing 1.0 and 0.5 clo clothing during primarily sedentary activity (≤1.1 met)].

ASHRAE Standard 55 specifies ranges for operative temperatures that will be acceptable to at least 90% of the occupants.



[2]

For example, for a sedentary activity level (70 W/m^2) and typical indoor clothing, the standard recommends the following **operative temperature ranges**:

20 to 24°C in winter (heating period, 0.14 to 0.155 m^2·K/W),

23 to 26°C in summer (cooling period, 0.08 m^2·K/W).

Operative temperatures **for activities higher than 70 W/m^2 (but less than 175 W/m2)** can be found from **ISO Standard 7730** or can be calculated from the operative temperatures at sedentary conditions using the following equation (ASHRAE Standard 55):

$$\boldsymbol{t_o^{act} = t_o^{sed} - 0.33(0.155 + R_{cl})(M - 70)}$$

where

- t_o^{act} = operative temperature for activity, °C
- t_o^{sed} = operative temperature for sedentary conditions, °C
- R_{cl} = insulation value for garment ensemble, m²·K/W (**1 clo = 0.155 m²·K/W**)
- M = metabolic rate, W/m²

Observations:

Table 13 Equivalent Wind Chill Temperatures of Cold Environments

Wind Speed, km/h	Actual Thermometer Reading, °C												
	10	5	0	–5	–10	–15	–20	–25	–30	–35	–40	–45	–50
	Equivalent Wind Chill Temperature, °C												
Calm	10	5	0	–5	–10	–15	–20	–25	–30	–35	–40	–45	–50
10	8	2	–3	–9	–14	–20	–25	–31	–37	–42	–48	–53	–59
20	3	–3	–10	–16	–23	–29	–35	–42	–48	–55	–61	–68	–74
30	1	–6	–13	–20	–27	–34	–42	–49	–56	–63	–70	–77	–84
40	–1	–8	–16	–23	–31	–38	–46	–53	–60	–68	–75	–83	–90
50	–2	–10	–18	–25	–33	–41	–48	–56	–64	–71	–79	–87	–94
60	–3	–11	–19	–27	–35	–42	–50	–58	–66	–74	–82	–90	–97
70	–4	–12	–20	–28	–35	–43	–51	–59	–67	–75	–83	–91	–99

Little danger: In less than 5 h, with dry skin. Maximum danger from false sense of security. (WCI < 1400)

Increasing danger: Danger of freezing exposed flesh within 1 min. (1400 ≤ WCI ≤ 2000)

Great danger: Flesh may freeze within 30 s. (WCI > 2000)

Source: U.S. Army Research Institute of Environmental Medicine.
Notes: Cooling power of environment expressed as an equivalent temperature under calm conditions [Equation (86)]. Winds greater than 70 km/h have little added chilling effect.

In the context of human-centered AI, building management systems utilize these **ASHRAE boundaries** not as static setpoints, but as a **dynamic mathematical polygon**. **Machine learning algorithms continuously adjust the HVAC outputs to let the temperature drift naturally within this acceptable zone**—taking advantage of passive solar gains or thermal inertia—optimizing for the lowest energy consumption while **keeping the Predicted Mean Vote (PMV) index as close to neutral as possible**.

Ventilation rate

Minimum ventilation rates for commercial and residential buildings are specified in ASHRAE Standards 62.1 and 62.2, respectively, and for health care facilities in ASHRAE Standard 170. New ASHRAE Standard 241, Control of Infectious Aerosols, establishes additional ventilation rates (equivalent clean air, not necessarily just outdoor air) that can be used for control of infectious aerosols in any type of facility during infection risk management mode (i.e., periods of perceived high risk of transmission).

Special attention should be cared about air draft.

A study of the effect of air velocity over the whole body found thermal acceptability unaffected in neutral environments by **air speeds of < 0.25 m/s** (Berglund and Fobelets 1987).

Obs: 1 km/h = 1000/3600 m/s = 0.27 m/s

SR EN 16798-1:

Over long periods, the ventilation rate of a used building must not fall below

14.4 m3/h/person

AND not less than **0.7 m^3/h/m^2** (for each m^2 of surface area of the building).

2.1.3 Visual comfort

Light is the most important environmental cue for synchronizing or "entraining" our **internal biological circadian rhythms** (e.g., sleep/wake, hormone release, appetite and digestion, body temperature, etc.) with the 24 h solar cycle.

Poor sleep has been linked to poor performance, increased anxiety and depression, and diminished overall health.

Various metrics have been proposed for measuring and specifying interior lighting for the circadian system.

Interior light levels required for visual tasks typically reach about 200 to 300 lux at the eyes, being vastly lower than the **circadian-effective level of >10,000 lux** received at the eyes in outdoor daylight, even on a cloudy day.

This has generally led to an indoor lighted environment that is visually adequate but deficient in light for stimulating the circadian system (Rea et al. 2010).

SR EN 12464-1 (Tabel 5.2):

Building categories	**Offices with writing, reading, data processing activities; Conference/meeting rooms**	**Educational buildings, libraries, auditoriums**	**Annexes to educational buildings (sports halls, swimming pools), clubs**
Illuminance level [**lux**= =lm/m²]	**500** Controllable lighting	**500** Controllable lighting	300 Lighting depending on the type of training

2.2. Thermal comfort indices

Methods for evaluating the general thermal state of the body both in comfort conditions and under heat and cold stress are based on an **analysis of the heat balance for the human body**.

A person may find the thermal environment unacceptable or intolerable due to local effects on the body caused by asymmetric radiation, air velocity, vertical air temperature differences, or contact with hot or cold surfaces (floors, machinery, tools, etc.).

Moderate Thermal Environments: **ISO Standard 7730** defines the **predicted mean vote (PMV) and predicted percent dissatisfied (PPD) indices** (Fanger 1982) for evaluating moderate thermal environments.

Predicted Mean Vote (PMV)

To quantify comfort, the **PMV index** gives a value on the seven-point **ASHRAE thermal sensation scale**:

+3 hot

+2 warm

+1 slightly warm

0 neutral

-1 slightly cool

-2 cool

-3 cold

An equation in the standard for calculating the PMV index is based on six factors: clothing, activity, air temperature, mean radiant temperature, air speed, and humidity.

Even if the PMV is 0, at least 5% of the occupants will be dissatisfied with the thermal environment.

The PMV index is determined assuming that all evaporation from the skin is transported through the clothing to the environment; therefore, the PMV index is applicable only within -2 < PMV < +2, that is, for thermal environments where sweating is minimal. **The PMV index is not applicable for hot environments**.

PMV is related to the imbalance between actual heat flow from the body in a given environment and the heat flow required for optimum comfort at the specified activity.

$$\mathbf{PMV} = (\mathbf{0.303\, e^{-0.036M}} + \mathbf{0.028})\boldsymbol{L}$$

L is the thermal load on the body, defined as the difference between internal heat production and heat loss to the actual environment (=*M - Human body thermal losses*).

Predicted Percent Dissatisfied (PPD)

$$\mathbf{PPD} = \mathbf{100} - \mathbf{95\ exp}[-(\mathbf{0.03353PMV^4} + \mathbf{0.2179PMV^2})]$$

Overall thermal comfort state (I5 – Romanian regulation "Normativ pentru proiectarea, executarea şi exploatarea instalaţiilor de ventilare şi climatizare")

Environmental category – targeted level	**PPD**	**PMV**
I - High level– children, elderly, sick individuals	**< 6%**	**-0.2 < PMV < 0.2**
II - Medium – considered normal in new/renovated buildings	**< 10%**	**-0.5 < PMV < 0.5**
III - Moderate – no health risk but lower comfort, in existing buildings	**< 15%**	**-0.7 < PMV < 0.7**
IV - Low – no health risk but low comfort – for limited periods	**> 15%**	**PMV < -0.7 or PMV > 0.7**

The **PMV-PPD model** is widely used and accepted for design and field assessment of comfort conditions. **ISO Standard 7730** includes a short computer listing that facilitates computing PMV and PPD for a wide range of parameters.

Algorithm for Calculating PMV and PPD Indices (Based on ISO 7730)

Step 1: Initialization of Environmental and Personal Variables

The algorithm requires six **primary input parameters** collected from sensors and occupant profiles:

- t_a: air temperature (°C)
- t_{mr}: mean radiant temperature (°C)
- v: air velocity (m/s)
- RH: relative humidity (%)
- I_{clo}: clothing insulation (clo)
- M_{met}: metabolic rate (met)

Step 2: Unit Conversions and Preliminary Calculations

The standard comfort models require standard metric units (W/m^2 for energy and m^2K/W for insulation).

- **Metabolic rate conversion:**

$$M = M_{met} \cdot 58.15$$

- **Clothing insulation conversion:**

$$I_{cl} = 0.155 \cdot I_{clo}$$

- **Partial water vapor pressure (p_v):** Calculated using the Antoine equation for saturated vapor pressure.

$$p_{sat}(t_a) = e^{16.6536 - \frac{4030.183}{t_a + 235}} \; [kPa]$$

$$p_v[Pa] = \frac{RH[\%]}{100} \cdot (p_{sat}(t_a) \cdot 1000)[Pa]$$

- **Clothing surface area factor (f_{cl})** = the ratio of the surface area of the clothed body to the surface area of the nude body.

$$f_{cl} = \begin{cases} 1 + 1.29 \cdot I_{cl}, & I_{cl} < 0.078 \\ 1.05 + 0.645 \cdot I_{cl}, & I_{cl} \geq 0.078 \end{cases}$$

Step 3: Iterative Calculation of Clothing Surface Temperature (t_{cl})

The temperature of the clothing surface cannot be solved directly; it is determined through a numerical iteration (successive approximation) balancing the heat conducted through the clothing with the heat lost to the environment via radiation and convection.

1. **Initial estimation:** A starting absolute temperature T_{cl} is approximated based on internal variables.
2. **Convective heat transfer coefficient (h_c):** The algorithm evaluates both forced convection ($h_{c,f}$) and natural convection ($h_{c,n}$) and selects the dominant heat transfer mode at each iteration:

- Forced convection:

$$h_{c,f} = 12.1 \cdot v^{0.5}$$

- Natural convection:

$$h_{c,n} = 2.38 \cdot |t_{cl} - t_a|^{0.25}$$

- Selection:

$$h_c = \max(h_{c,f}, h_{c,n})$$

3. **Iteration:** The value of t_{cl} is recalculated until the difference between successive iterations is less than a threshold error ($\epsilon = 0.0015$).

Step 4: Human Body Heat Loss Components

Once t_{cl} and h_c converge, the algorithm calculates the six distinct paths of heat loss from the human body to the environment:

1. **Skin diffusion loss (HL_1):** Moisture diffusing through the skin

$$HL_1 = 3.05 \cdot 10^{-3} \cdot (5733 - 6.99 \cdot M - p_v)$$

2. **Sweat loss (HL_2):** Evaporative heat loss from regulatory sweating (only if $M > 58.15\ W/m^2$)

$$HL_2 = 0.42 \cdot (M - 58.15)$$

3. **Latent respiration loss (HL_3):** Heat lost through moisture exhaled from the lungs

$$HL_3 = 1.7 \cdot 10^{-5} \cdot M \cdot (5867 - p_v)$$

4. **Dry respiration loss (HL_4):** Sensible heat lost heating the inhaled air

$$HL_4 = 0.0014 \cdot M \cdot (34 - t_a)$$

5. **Radiation loss (HL_5):** Heat radiated from the clothing surface to the surrounding room surfaces (temperatures are in Kelvin: T_{cl} and T_{mr})

$$HL_5 = 3.96 \cdot f_{cl} \cdot \left[\left(\frac{T_{cl}}{100}\right)^4 - \left(\frac{T_{mr}}{100}\right)^4\right]$$

6. **Convection loss (HL_6):** Heat convected from the clothing surface to the room air

$$HL_6 = f_{cl} \cdot h_c \cdot (t_{cl} - t_a)$$

Step 5: Calculation of PMV and PPD

The thermal sensation transfer coefficient (TS) correlates the internal metabolic heat production to the skin's thermal sensitivity

$$TS = 0.303 \cdot e^{-0.036 \cdot M} + 0.028$$

Net thermal load on the body (internal heat production minus all heat losses):

$$L = M - HL_1 - HL_2 - HL_3 - HL_4 - HL_5 - HL_6$$

The **Predicted Mean Vote (PMV)** is calculated by:

$$PMV = TS \cdot L$$

The **Predicted Percentage Dissatisfied (PPD)** is derived statistically from the PMV index:

$$PPD = 100 - 95 \cdot e^{-0.03353 \cdot PMV^4 - 0.2179 \cdot PMV^2}$$

Step 6: Calculation of Operative Temperature (t_o)

The algorithm applies an air-velocity dependent weighting factor to calculate the operative temperature, which the AI can use as a primary control variable:

$$t_o = \begin{cases} 0.5 \cdot t_a + 0.5 \cdot t_{mr}, & v < 0.2\ m/s \\ 0.6 \cdot t_a + 0.4 \cdot t_{mr}, & 0.2\ \frac{m}{s} \leq v < 0.6 m/s \\ 0.7 \cdot t_a + 0.3 \cdot t_{mr}, & v \geq 0.6\ m/s \end{cases}$$



LECTURE CHALLENGE – write the code in PYTHON for computing *PMV*, *PPD* and *to* for the considered day in the previous simulation (exterior air parameters).
Save it as calculate_pmv_ppd(t_a, t_mr, v=, rh=, i_clo, m_met).py
Results should appear in the form: 'PMV=', 'PPD=', 't_o=', 'n iterations='.

Then, find interior comfort conditions so that -0.5 < PMV < 0.5 (change t_a, t_{mr}, *RH*).

Summer day in Romania (exterior air)		**Interior air conditions**	
t_a = 30°C	v = 0.15 m/s	t_a = °C	v = 0.15 m/s
t_{mr} = 30.5°C	clo = 0.5	t_{mr} =°C	clo = 0.5
RH=φ = 60%	M_{met} =1.2 met	RH=φ = 60%	M_{met} =1.2 met
PMV = 1.767		PMV =	
PPD = 65.3		PPD =	
t_o = 30.25 °C		t_o = °C	
n iterations = 8		n iterations =	

2.3. Reference standards and regulations

Standards describe consistent methods of testing, specify confirmed design guidelines, and recommend standard practices.

ASHRAE Standards become the vital document to assess the indoor air quality:

- ANSI/ASHRAE Standard 55-1992, Thermal Environmental Conditions for Human Occupancy
- ASHRAE/IES Standard 90.1-1999, Energy Standard for Buildings Except Low-Rise Residential Buildings, which is often called the Energy Standard. This standard is under continuous maintenance by a standing standard committee (SSPC) for regular publication of addenda or revisions.
- ASHRAE Standard 62-1999, Ventilation for Acceptable Indoor Air Quality, which is often called the Indoor Air Quality Standard. This standard is under continuous maintenance.
- ASHRAE Standard 15-1992, Safety Code for Mechanical Refrigeration.
- ANSI/ASHRAE Standard 34-1997, Number Designation and Safety Classification of Refrigerants. This standard is under continuous maintenance.
- ANSI/ASHRAE Standard 135-1995, BACnet: A Data Communication Protocol for Building Automation and Control Networks.
- Air Conditioning and Refrigeration Institute, Standards for Unitary Equipment.

- ISO 9000 Guidelines of the Selection and the Application of Quality Management and Quality Assurance Standards

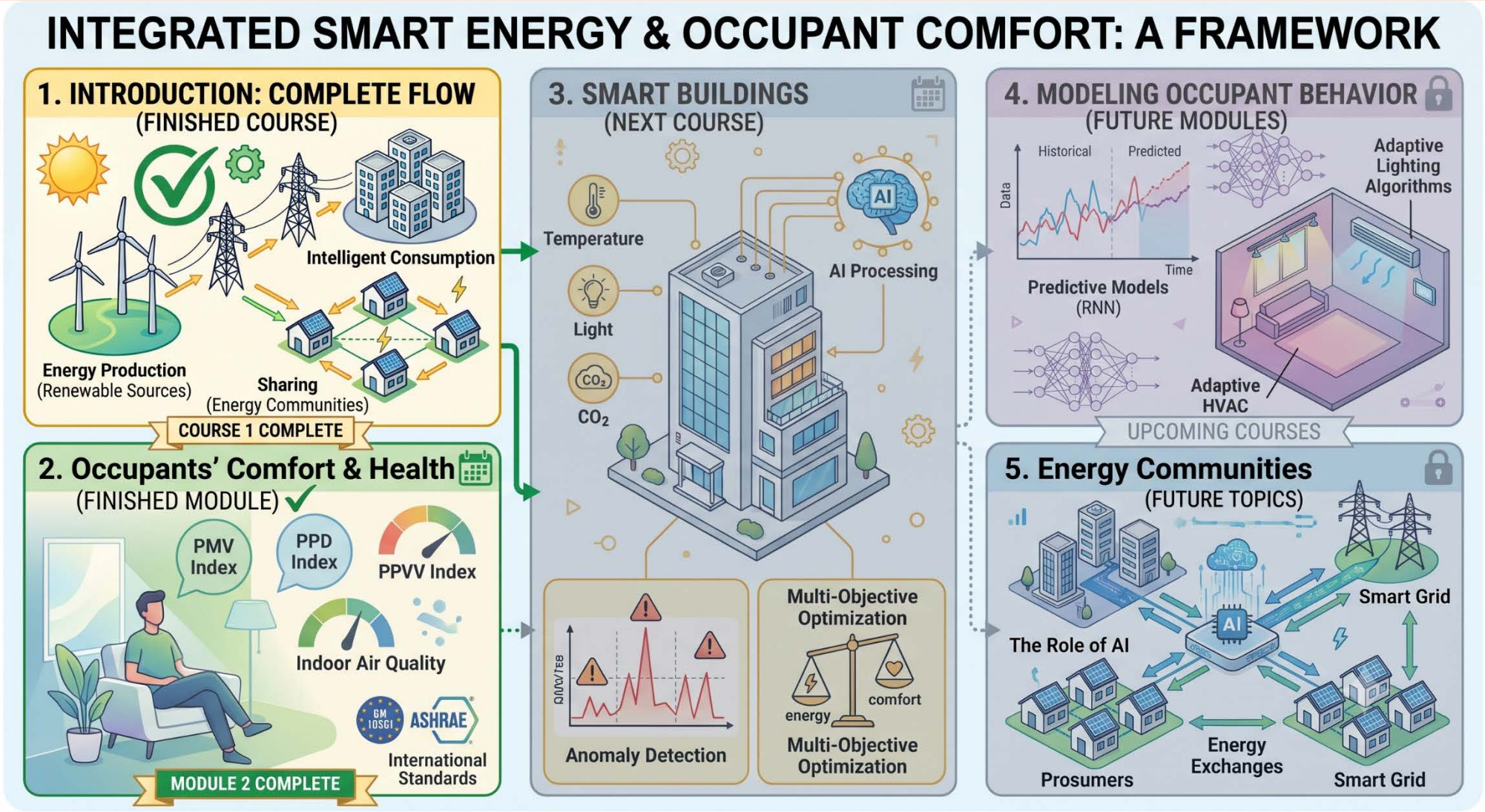
- ISO 9001 Modeling of Quality Assurance in Design Development, Manufacturing, Installation and Servicing
- ISO 9002 Modeling of Quality Assurance in Production and Installation
- ISO 9003 Modeling of Quality Assurance in Final Inspection and Testing
- ISO 9004 Guidelines for the Implementation of Quality Management and to Provide Quality Systems

References

[1] Shan K. Wang, Handbook of air conditioning and refrigeration, 2nd ed. McGraw-Hill 2000.

[2] ASHARE Handbook Fundamentals, 2025

UPCOMING COURSES



Chapter 3 – SMART BUILDINGS

- Fundamentals of modeling and optimizing energy systems to maintain thermal comfort parameters.
- Integration of temperature, humidity, presence data, (anonymized) video streams and CO2 sensors.
- Using algorithms to detect faulty sensors without human intervention.
- Multi-objective optimization (simultaneously reducing the bill cost, minimizing the carbon footprint and maintaining thermal comfort).
- AI application areas in buildings.

Chapter 4 – MODELING AND PREDICTING OCCUPANT BEHAVIOR

- Using recurrent neural networks (RNN) to learn occupants' routine and condition (pre-heat/cool) spaces before they start their activity.
- Estimating the number of occupants and correlating with individualized thermal demand: metabolic rate per activity type according to ISO 7730; aggregation of individual demands at zone and building level.
- Integration of mobile applications through which occupants vote on the comfort level, to train learning models.
- Intelligent Control of HVAC and Lighting Systems: using weather forecasts and dynamic energy prices to store thermal energy.
- Algorithms that combine natural light with artificial light by controlling blinds, etc.
- Ethical and acceptability aspects: occupant data privacy (GDPR), explainability of system decisions to the user, automated decision criteria vs. human intervention.

Chapter 5 – ENERGY COMMUNITIES

- The role of AI in facilitating energy exchanges between prosumers.
- Applications in Smart Grids – data analysis and suggesting integration / component placement solutions.